Supporting Information:

Silent but deadly? Physics says rarely, memory says always

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Contents: the chain from release to memory (Figure S1); CFD validation (Section S1); the CFD scenes in detail (Sections S2–S5); breath sampling (Section S6); the cross-cultural survey (Section S7); and the full parameter table (Section S8). File paths in square brackets point into the code and data archive.

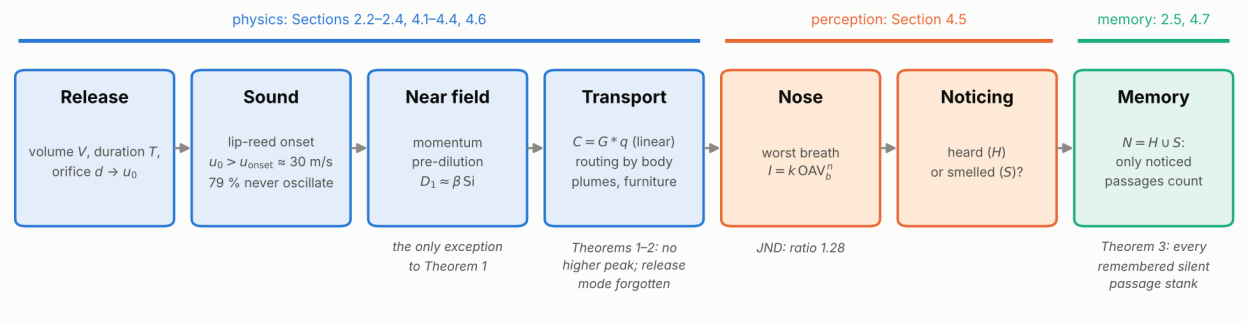


Figure S1: The chain from release to memory, the quantity each stage hands on, and where Theorems 1–3 act.

S1 CFD validation

[cfd/validation/m3_report.md]

- *Free round jet* (axisymmetric, Re = 6700, three meshes). This gives a velocity-decay constant of 5.0 and a scalar constant $K_c = 4.3$–4.6, with grid uncertainty of at most 3.7%. The known round-jet anomaly of SST (about 25% excess spreading) means the model over-dilutes audible jets by about 15%, which favours the silent release.

- *Convective boundary layer* of a simplified seated manikin with skin at 33 °C, compared with thermal-manikin data.[S1] The face-front peak velocity is 40–56% too high at room temperatures of 20–26 °C, while the trend with room temperature is reproduced (-27% vs -33% measured). The fast boundary layer shortens travel times to the releaser's own nose by about 30%; the analytic model compensates by stretching the CFD times by a factor of 1.4.

S2 CFD scenes

All scenes use the solver of the main text (transient RANS k–ω SST, Boussinesq, internal-energy form), skin at 33 °C and walls at 23 °C. Manikins are built from geometric primitives. The release is a momentum-equivalent inlet behind the releaser's buttocks, pointing down and back: silent, 92 mL over 2 s through 8 mm ($u_0 = 0.9\,\mathrm{m\,s^{-1}}$); audible, 92 mL over 0.5 s through 2.4 mm ($41\,\mathrm{m\,s^{-1}}$). The odorant is a passive scalar, sampled at probes 2.5 cm in front of each nose, on 5 cm breathing-zone discs at nose height and over sinusoidal breathing cycles. Each scene first runs a precursor that develops the body plumes; the silent and audible branches then start from the same field.

- *Sofa duet.* A 3.2 × 3.2 × 2.6 m living room at 23 °C with a 0.5 m ceiling vent for pressure relief [ASSUMPTION]. Two seated manikins sit 0.6 m apart on a two-seat sofa whose base is solid to the floor. Mesh 0.72 million cells, refined to 5 mm at the release and 2.5 mm on the inlet face. Precursor 30 s; branches 30 s, the upright one extended to 120 s. Two posture variants rotate the releaser's upper body about the

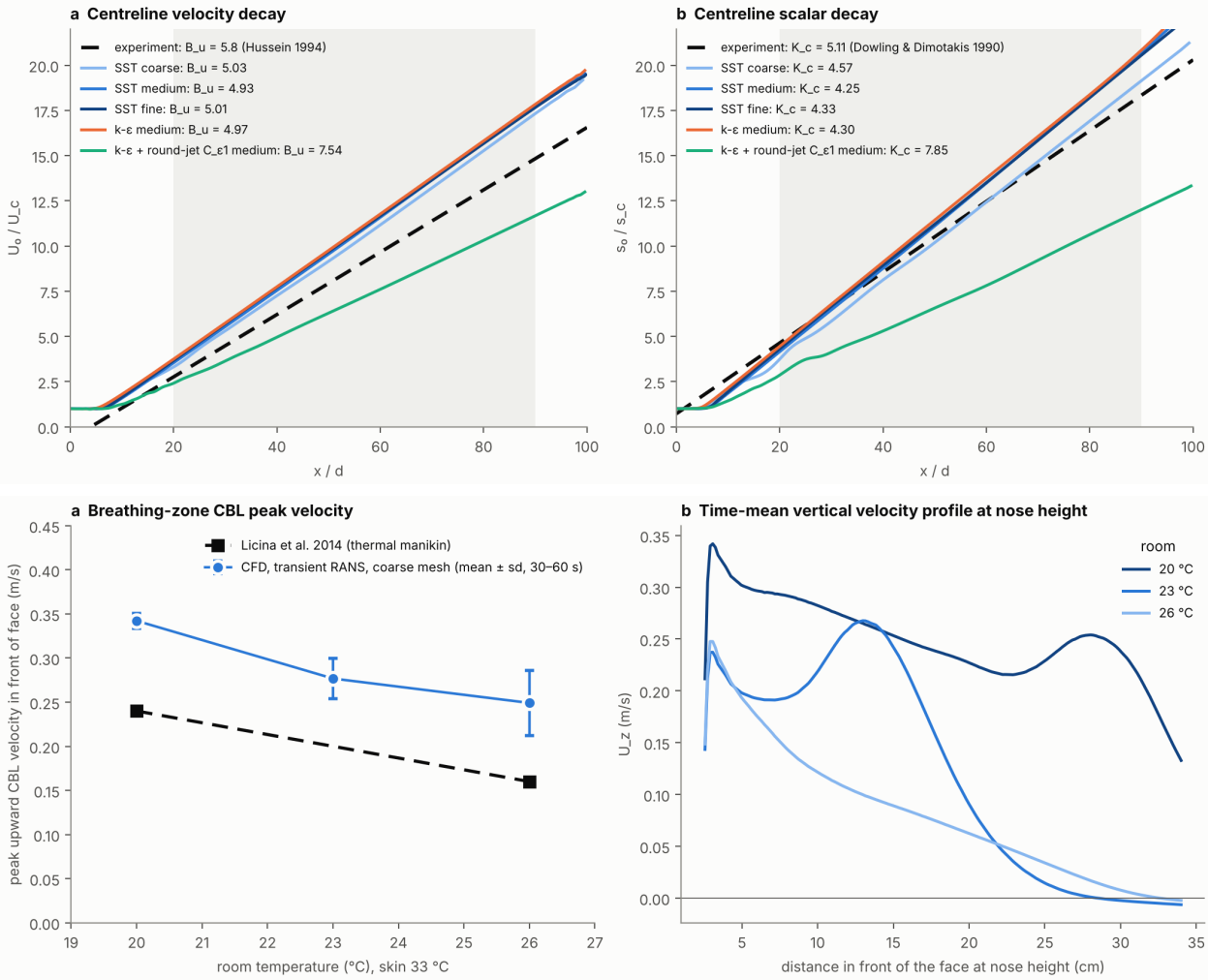


Figure S2: Validation. Top: free round jet at Re = 6700 on three meshes (centreline velocity and scalar decay, spreading). Bottom: face-front convective boundary layer of the seated manikin against thermal-manikin data[S1] at 20–26 °C, and time-mean vertical velocity profiles at nose height.

hip: reclined by 4.5 °, so that the back closes onto the backrest, and leaning forward by 10 °, so that the gap behind the back opens [cfd/cases/sofa/m4_report.md].

- *Elevator.* The car of a 630 kg, eight-person elevator, 1.10 m wide, 1.40 m deep and 2.10 m high (a manufacturer's standard car, within the maximum area of 1.66 m^2 allowed for that load). Four standing manikins (nose at 1.59 m) stand in a 2×2 grid facing the door; the fan is off and a 0.2 m ceiling opening relieves pressure [ASSUMPTION]. Mesh 0.91 million cells. Precursor 30 s; four 60 s branches release behind a front-row or a back-row occupant, and mirror symmetry gives the other two positions [cfd/cases/elevator/m5_report.md].

- *Standing pair.* The living room of the sofa duet with two standing manikins face to face, noses 1.0 m apart, and 0.8 m of open floor behind each back. Two air states share one mesh (0.43 million cells): still air, and office ventilation through one square ceiling diffuser above the pair, 21.2 L s^{-1} = 2.9 air changes per hour (EN 16798-1 Category II for two people in a low-polluting 10 m^2 room: 7 L s^{-1} per person plus 0.7 L s^{-1} m^{-2} [S2]), discharged horizontally along the ceiling at 1.5 m s^{-1} [ASSUMPTION] at room temperature, with a ceiling exhaust grille. Precursors 30 s (still) and 90 s (ventilated); 60 s branches [cfd/cases/standing].

S3 Sofa duet

Silent release. The silent release has too little momentum to leave the convective boundary layer of the releaser's back. It is drawn up the gap between back and backrest, over the head and on to the ceiling. Within 30 s its breathing-zone OAV stays at or below 10^{-10} at both noses, while the region within 25 cm of the releaser's head reaches an OAV of 6 000.

Audible release. The audible jet splashes into the corner between seat and backrest, spreads sideways along the gap and is lifted up both backs as a wider, more dilute sheet; part of it curls around the heads. It reaches the neighbour's head region within 6 s and both

breathing zones at OAV 0.2–0.6.

Reclined posture. Reclining the releaser by 4.5 ° closes the back onto the backrest. The silent release then leaves the seat-level wedge sideways and still rises behind the head, so no breathing zone receives a detectable dose. The audible jet now wraps around the releaser's face, exceeding threshold at their own nose (OAV 2–4).

Forward lean. Leaning the releaser forward by 10 ° opens the gap behind the back. Both releases then rise behind the head, and the releaser's nose, now 13 cm further forward, receives neither (breathing-zone OAV $\leq 10^{-5}$). The silent cloud reaches the region within 25 cm of the neighbour's head as strongly as the audible one (OAV 580 against 450) but not the neighbour's breathing zone (10^{-4}); the audible jet reaches it at OAV 2.6.

Longer window. Extending the upright case to 120 s shows what happens after the direct pass. Both clouds reach the ceiling, come back down through the room and arrive at both noses from about 45 s on. After about 70 s the silent and audible traces at the neighbour are indistinguishable (worst breath 0.39 for both), as Theorem 2 requires. At the releaser's own nose the mixed silent cloud runs 1.5–2 times higher and touches the trained-panel threshold for about a second near 100 s. These late levels are 10–23 times below the thresholds of an untrained observer, and they arrive when the release can no longer be attributed to anyone.

Table S1: Sofa duet: peak breathing-zone OAV within 30 s of release onset (RANS mean field).

nose	posture	silent	audible
releaser	upright	5×10^{-11}	0.46
releaser	reclined 4.5 °	9×10^{-11}	3.3
releaser	forward 10 °	3×10^{-7}	1.5×10^{-5}
neighbour	upright	3×10^{-14}	0.63
neighbour	reclined 4.5 °	3×10^{-15}	0.19
neighbour	forward 10 °	1×10^{-4}	2.6

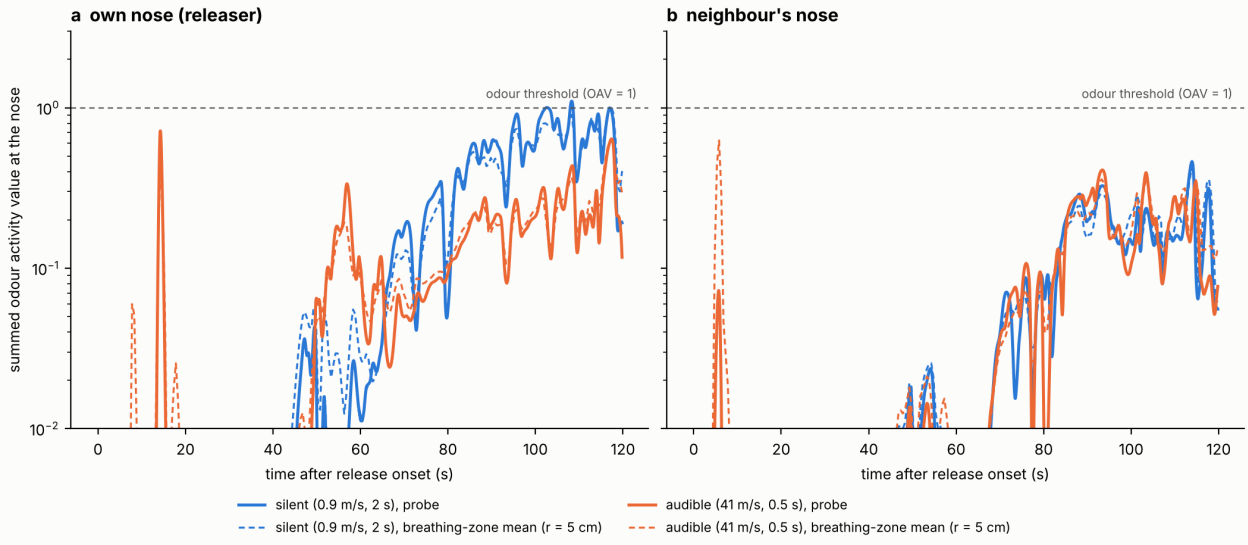


Figure S3: Sofa duet, upright posture: OAV at the releaser's and the neighbour's nose over 120 s for the same 92 mL released silently or audibly. During the direct pass (< 45 s) only the audible release reaches the noses; after room recirculation both modes converge. Solid: probe 2.5 cm in front of the nose; dashed: mean over a 5 cm breathing-zone disc. The horizontal line is the odour threshold (trained-panel thresholds).

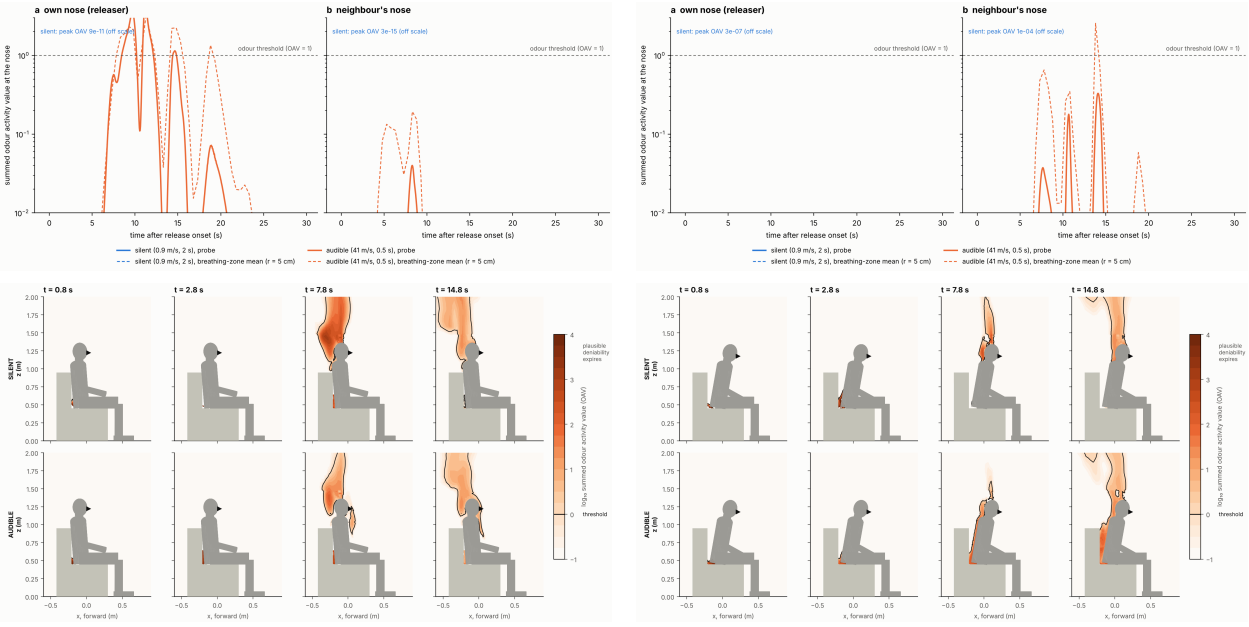


Figure S4: Sofa duet, posture variants over 30 s: reclined by 4.5 ° (left) and leaning forward by 10 ° (right). Top: OAV at both noses, as in Figure S3. Bottom: $\log_{10}$ OAV in the releaser's mid-sagittal plane, silent (upper rows) and audible (lower rows).

S4 Elevator

What stands behind the culprit decides. A silent release follows the culprit's own back plume.

- With a person behind (front-row culprit), it rises in the 0.2 m gap between the culprit's back and that person's face, straight into their breathing zone. The silent release is stronger at every nose ($R = 1.4$–4.3), and strongest, OAV 27, for the person behind.
- With the car wall behind (back-row culprit), it rises in the gap between back and wall, the chimney of the sofa, and reaches the front row only below threshold. The audible jet, which hits the floor and is lifted by all four body plumes, is now stronger at every nose ($R = 0.20$–0.55).

Who smelt it. With a front-row culprit, the person behind smells it first (8.8 s), before the culprit (17.6 s). The folk rule that the first to smell it dealt it is therefore right only for a back-row culprit, i.e. in half of the positions, for both modes.

Can the others tell? Each occupant other than the culprit reacts with the detection probability of the main text's observer models, independently; observers see only who reacted and infer the culprit with a uniform prior, for a culprit who stays still and one who reacts along. An everyday, distracted car notices about nine passages in ten not at all. When someone does react, the culprit is identified barely above chance (0.28–0.31, against 0.25). Attentive occupants do better, but only about one time in two (0.39–0.52). Odour gives no better identification for silent passages than for audible ones; the audible passage also carries the culprit's acoustic signature, which is not modelled.

S5 Standing pair

Results pending (simulations running).

Table S2: Elevator: expected worst-breath OAV within 60 s at each occupant's nose (trained-panel thresholds; RANS mean field), for a front-row and a back-row culprit.

culprit	nose	silent	audible	R
front row	culprit	6.7	1.9	3.4
	beside	3.6	0.8	4.3
	behind	27	20	1.4
	diagonal	7.6	2.7	2.9
back row	culprit	4.5	23	0.20
	beside	2.3	5.9	0.39
	in front	0.57	2.8	0.20
	diagonal	0.41	0.74	0.55

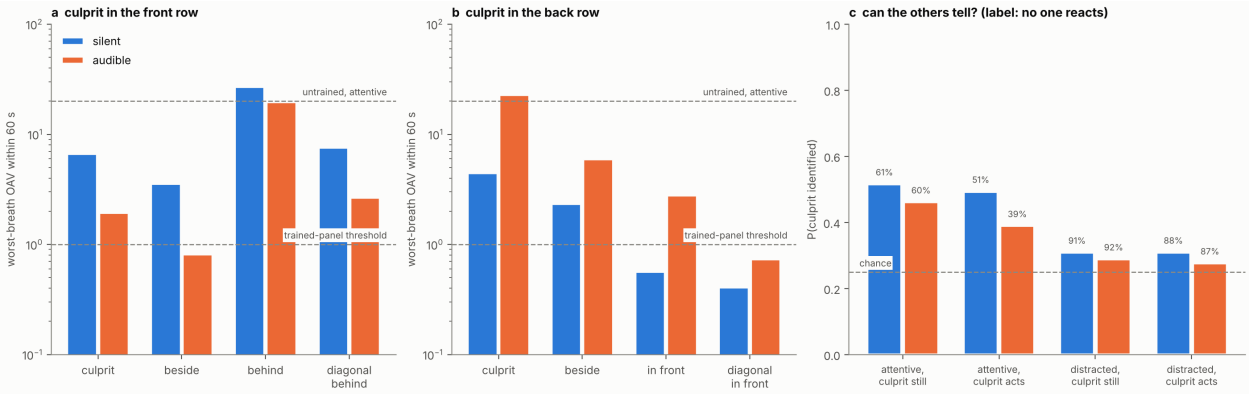


Figure S5: Elevator whodunit: four standing occupants in a closed 630 kg car, fan off, same 92 mL. (a, b) Expected worst-breath OAV within 60 s at each nose for a front-row and a back-row culprit; dashed lines mark the trained-panel threshold and the attentive population threshold. (c) Probability that the others identify the culprit from who reacted, for attentive and distracted onlookers and a culprit who stays still or reacts along; the labels give the probability that no one reacts. The dashed line is chance.

S6 Breath sampling

An audible puff reaches the neighbour after a median of 16 s and passes with a time scale of about 3.8 s (10–90%: 1.4–12 s). Unwarned, the breathing phase barely helps: for a median passage the worst breath still captures 81–99% of the peak. A listener warned by the sound needs about 23 s of breath-hold for an even chance of missing the puff (worst breath below 10% of the peak); 33 s, the typical maximum at end-expiration, gives 68% [perception/m6_report.md].

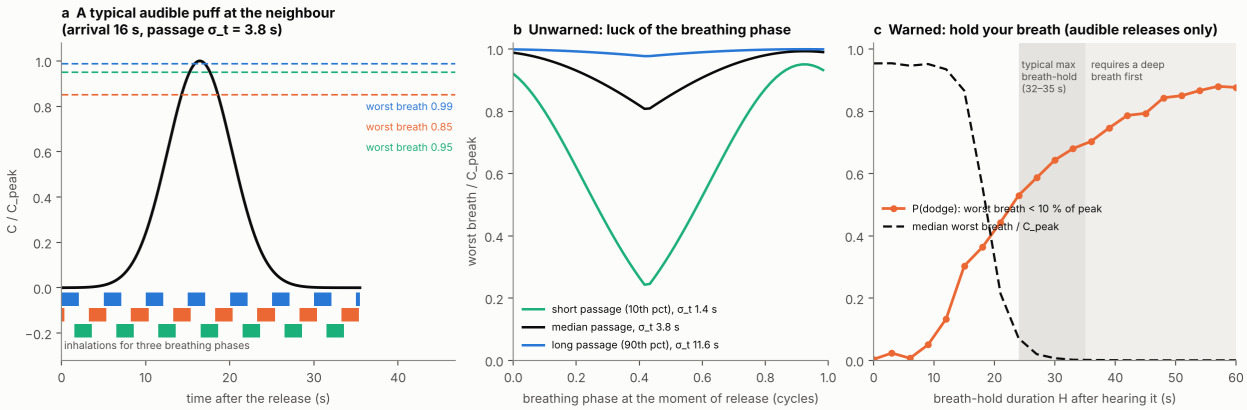


Figure S6: Breath sampling at the neighbour's nose for audible releases. (a) A typical puff (arrival 16 s, passage time scale 3.8 s) and the inhalations of three breathing phases, with the worst breath of each. (b) Unwarned: worst breath relative to the peak against the breathing phase at release, for short, median and long passages. (c) Warned: probability of dodging the puff (worst breath below 10% of the peak) and median worst breath, against the breath-hold duration after hearing the release (0.5 s reaction latency).

S7 Cross-cultural survey

We searched dictionaries, proverb collections and etymological handbooks in 32 languages or historical stages for a dedicated word for a silent passage and for an explicit claim that it smells worse; every entry used was checked against a fetched source [culture/survey_summary.md, culture/cross_cultura

Each language is graded by its strongest evidence: A, an explicit saying (traditional, or in media or film only); B, a dedicated word defined by its smell; C, a dedicated word without a

smell claim; D, nothing found. The silent/loud distinction is lexicalised in at least five language families; the explicit claim is traditional only in Chinese and English, and smell-defined silent words arose independently at least four times within Indo-European. Documentary presence is not prevalence.

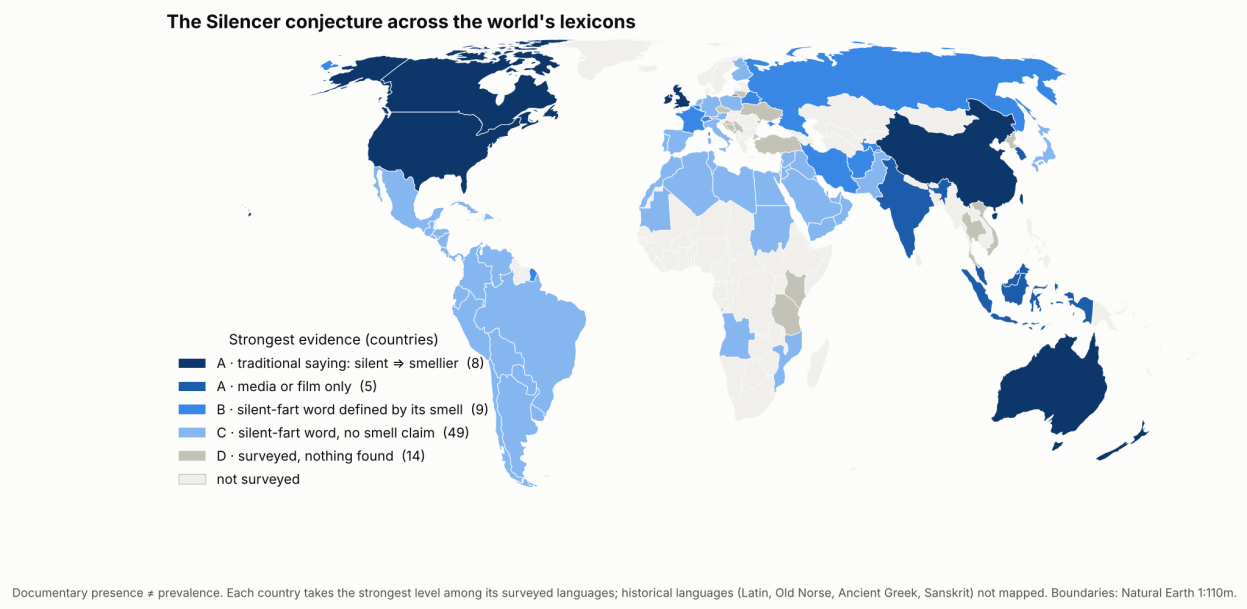


Figure S7: The Silencer conjecture across languages, by evidence level. Each country takes the strongest level among its surveyed languages. A: an explicit saying that silent passages smell worse (traditional, or in media or film only); B: a dedicated word for a silent passage defined by its smell; C: a dedicated word with no smell claim; D: surveyed, nothing found.

S8 Parameters

All parameters, with their sources, ranges and status (verified, derived, partial or assumption), are generated from one file `[params/params.yaml]`.

Table S3: Model parameters: nominal value, range sampled or tested, and status (VERIFIED, DERIVED, PARTIAL, ASSUMPTION). Sources and notes for every entry are in `params/params.yaml` of the code archive.

parameter	nominal	range	units	status
Source				
`T_gas`	310.15	309.85–310.65	K	VERIFIED
`V_passage`	9.2×10^{-5}	1.7×10^{-5}–3.75×10^{-4}	m^3	VERIFIED
`V_daily`	7.05×10^{-4}	4.76×10^{-4}–0.001491	$m^3\,d^{-1}$	VERIFIED
`n_daily`	10	1–20	d^{-1}	VERIFIED
`d_orifice`	0.004	0.001–0.015	m	ASSUMPTION
`T_release`	1	0.3–5	s	ASSUMPTION
`dp_rectal`	2000	500–4000	Pa	ASSUMPTION
`p_anal_rest`	8666	6133–12000	Pa	VERIFIED
`g_reduced`	1.7	-0.66–4.68	$m\,s^{-2}$	DERIVED
`nu_gas`	1.7×10^{-5}		$m^2\,s^{-1}$	ASSUMPTION
`c_H2S`	1.06	0.26–1.86	$\mu mol\,L^{-1}$	VERIFIED
`c_CH3SH`	0.21	0.05–0.37	$\mu mol\,L^{-1}$	VERIFIED
`c_DMS`	0.08	0.04–0.12	$\mu mol\,L^{-1}$	VERIFIED
`sulfur_sigma_ln`	1.1	1–1.25	–	DERIVED
Odour thresholds				
`H2S`	0.41	0.41–0.9	ppbv	VERIFIED
`CH3SH`	0.07		ppbv	VERIFIED
`DMS`	3		ppbv	VERIFIED
Jet and puff				
`B_u`	5.8	5.8–5.9	–	VERIFIED
`S_jet`	0.094	0.094–0.102	–	VERIFIED
`K_c`	5.11	4.73–5.2	–	VERIFIED
`Gamma_sj`	2.8	2.74–2.86	–	VERIFIED
`C_puff`	2.6	2.54–2.66	–	VERIFIED

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parameter	nominal	range	units	status
`alpha_th`	0.25	0.15–0.35	–	VERIFIED
`alpha_jet`	0.07	0.065–0.08	–	VERIFIED
`alpha_plume`	0.12	0.1–0.16	–	VERIFIED
Environment				
`u_handoff`	0.15	0.05–0.25	$\mathrm{m\,s^{-1}}$	ASSUMPTION
`u_CBL`	0.17	0.11–0.24	$\mathrm{m\,s^{-1}}$	VERIFIED
`u_CBL_pelvis`	0.1	0.05–0.15	$\mathrm{m\,s^{-1}}$	ASSUMPTION
`u_own_eff`	0.11	0.07–0.16	$\mathrm{m\,s^{-1}}$	DERIVED
`delta_CBL`	0.1	0.05–0.15	m	ASSUMPTION
`u_room`	0.08	0.03–0.15	$\mathrm{m\,s^{-1}}$	ASSUMPTION
`K_turb`	0.002	1×10^{-4}–0.013	$\mathrm{m^2\,s^{-1}}$	VERIFIED
`D_H2S_air`	1.65×10^{-5}	1.62×10^{-5}–1.77×10^{-5}	$\mathrm{m^2\,s^{-1}}$	PARTIAL
Geometry				
`eye_sit_m`	0.804	0.75–0.86	m	VERIFIED
`eye_sit_f`	0.748	0.698–0.799	m	VERIFIED
`eye_stand_m`	1.64	1.536–1.753	m	DERIVED
`eye_stand_f`	1.517	1.421–1.626	m	DERIVED
`L_own`	0.74	0.65–0.85	m	ASSUMPTION
`L_neighbour`	1	0.8–1.4	m	ASSUMPTION
Breathing				
`V_tidal`	6.3×10^{-4}	4×10^{-4}–8.3×10^{-4}	$\mathrm{m^3}$	VERIFIED
`f_breath`	12	10–16	$\mathrm{min^{-1}}$	VERIFIED
`Ti_Ttot`	0.42	0.34–0.5	–	VERIFIED
`T_insp`	2.1	1.6–2.7	s	DERIVED
Acoustics				

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parameter	nominal	range	units	status
`f0_flatus`	160	100–300	Hz	NON-PEER-REVIEWE
`bg_library`	40	35–45	dB(A)	ASSUMPTION
`bg_living`	35	30–45	dB(A)	ASSUMPTION
`bg_office`	45	43–55	dB(A)	VERIFIED
`bg_restaurant`	74	67–83	dB(A)	VERIFIED
`bg_subway`	77	74–80	dB(A)	VERIFIED
`P_th`	600	200–1300	Pa	ASSUMPTION
`C_d`	0.85	0.6–1	–	PARTIAL
`alpha_mod`	0.7	0.5–1	–	ASSUMPTION
`harm_slope`	-12	-18—6	dB/octave	ASSUMPTION
`n_harm`	8		–	ASSUMPTION
`A_cloth`	1	0–3	dB	PARTIAL
`r_listener`	0.6	0.4–3	m	ASSUMPTION
Perception and observers				
`n_stevens`	0.28	0.2–0.45	–	VERIFIED
`k_w_H2S`	0.95	0.8–1.2	intensity units per decade	PARTIAL
`weber_C`	0.28	0.05–0.4	$\Delta C/C$	VERIFIED
`tau_adapt`	80	40–120	s	ASSUMPTION
`a_inf`	0.35	0.25–0.4	–	PARTIAL
`sniff_ratio`	0.64	0.61–0.75	–	VERIFIED
`t_react`	0.5	0.3–1.5	s	ASSUMPTION
`H_max`	33	24–90	s	VERIFIED
`thr_scale_typ`	0.05	0.05–0.1	–	VERIFIED
`thr_sd_bs`	2		binary steps	VERIFIED
`distracted_factor`	26		–	VERIFIED

Table S3: Model parameters: nominal value, range sampled or tested, and status (VERIFIED, DERIVED, PARTIAL, ASSUMPTION). Sources and notes for every entry are in `params/params.yaml` of the code archive.

parameter	nominal	range	units	status
`distracted_sd_bs`	3.5		binary steps	VERIFIED
`recovery_gain`	1.5	1–2	–	ASSUMPTION

References

(S1) Licina, D.; Pantelic, J.; Melikov, A.; Sekhar, C.; Tham, K. W. Experimental investigation of the human convective boundary layer in a quiescent indoor environment. *Building and Environment* **2014**, *75*, 79–91.

(S2) Olesen, B. W. European standardisation on indoor air quality. *REHVA Journal* **2024**, 12–16.